

FIRST LAW OF THERMODYNAMICS (10 Marks)

Definition: Heat supplied to a system is used to increase internal energy and do external work.

Formula: $\delta Q = dU + \delta W$.

Statement:

Energy can neither be created nor destroyed.

Applications:

- Isothermal: $\Delta U=0$, $Q=W$
- Isochoric: $W=0$, $Q=\Delta U$
- Isobaric: $W=P\Delta V$
- Adiabatic: $Q=0$

Important points:

- Based on conservation of energy.
- Does not tell direction of heat flow.

SECOND LAW OF THERMODYNAMICS (10 Marks)

Kelvin–Planck: No engine can convert all heat into work.

Clausius: Heat cannot flow from cold to hot without external work.

Entropy:

$dS=\delta Q_{\text{rev}}/T$

Importance:

- Explains direction of heat flow.
- 100% efficient engine impossible.
- Basis of refrigerators and heat engines.

ENTROPY (10 Marks)

Entropy is the measure of disorder.

Formula: $\Delta S=\int \delta Q_{\text{rev}}/T$

Reversible process:

Entropy of universe remains constant.

Irreversible process:

Entropy of universe increases.

Significance:

Predicts spontaneity and direction of natural processes.

ISOTHERMAL & ADIABATIC (10 Marks)

Isothermal:

Temperature constant.

$PV=\text{constant}$.

Work: $W=nRT \ln(V_2/V_1)$

Adiabatic:

No heat exchange.

$PV^\gamma=\text{constant}$.

Work: $W=(P_1V_1-P_2V_2)/(\gamma-1)$

Differences:

Isothermal: $Q\neq 0$, $\Delta U=0$

Adiabatic: $Q=0$, $\Delta U\neq 0$

EQUIPARTITION OF ENERGY (10 Marks)

Each degree of freedom possesses average energy $\frac{1}{2}kT$.

For one mole:

$$U = fRT/2$$

$$C_v = fR/2$$

$$C_p = C_v + R$$

$$\gamma = C_p/C_v$$

Monoatomic gas:

$$C_v = 3R/2$$

$$C_p = 5R/2$$

$$\gamma = 5/3$$

PLANCK'S LAW & BLACK BODY (10 Marks)

Black body absorbs and emits perfectly.

Planck assumed energy is quantized:

$$E = nh\nu$$

Explains black body spectrum.

Leads to:

- Wien's law
- Stefan-Boltzmann law

Remember:

Good absorbers are good emitters.

MAXWELL-BOLTZMANN / BOSE-EINSTEIN / FERMİ-DIRAC (10 Marks)

MB: Classical, distinguishable particles.

BE: Bosons, multiple particles in same state.

FD: Fermions, obey Pauli exclusion.

Examples:

MB → Gas molecules

BE → Photons

FD → Electrons

CARNOT ENGINE & REVISION

Efficiency:

$$\eta = 1 - T_2/T_1$$

Carnot theorem:

No engine is more efficient than a Carnot engine.

1-Mark:

$$C_p - C_v = R$$

$$PV = nRT$$

$$\text{Mean free path } \lambda = 1/(\sqrt{2}n\sigma)$$

Boson, White radiation,

Thermodynamic potentials,

Wien law.